Tree Propagation

Motivation

Naturalists ask a futuristic question: how would the trees of Vienna spread if nobody took care of them any more? We assume that all people have left the city and that seeds can establish everywhere. Trees reproduce only through seeds, which are carried by the wind or by animals. Over decades, fast-growing pioneer species colonise open land first and are later replaced by slow-growing, shade-tolerant species. A simulation model can show how today's street and park trees of a district would shape the future urban forest, and which species would prevail.

Model

Initialisation

The area is a grid of square cells with a specific edge length. Every cell is either empty or holds exactly one tree. A tree is an agent with two attributes: its species σ (one of the eleven species in Table 1) and its age a in years. Its height is described by a height class $k \in \{0, 1, \ldots, 8\}$ in steps of 5 m (class 1: 0–5 m, class 2: 6–10 m, …, class 8: more than 35 m). A tree climbs one class every T_σ years until it reaches the maximal class K_σ of its species:

$$k = \min\big(\lfloor a/T_\sigma \rfloor,\ K_\sigma\big). \tag{1}$$

Initial trees are either placed on randomly chosen distinct cells (Tasks 2–4) or taken from the Vienna tree register *Baumkataster* (Task 5). If only the height class k of a tree is known, its age is set to $a = k\,T_\sigma$.

Dynamics

One time step corresponds to one year. All trees are updated simultaneously. Seed production and dispersal use the state at the beginning of the year. Every year consists of five steps.

1. Seed production. A tree is mature if its age reaches the age at maturity, $a \geq A_\sigma$. A mature tree produces

$$N \sim \text{Poisson}(\varphi_\sigma\, k) \tag{2}$$

effective seeds (seeds that are able to germinate), where φ_σ is the fecundity factor of the species. Taller trees thus produce more seeds.

2. Seed dispersal. Every seed flies an exponentially distributed distance r with mean $\bar{r}$, i.e. $r \sim$ Exp with $\mathbb{E}[r] = \bar{r}$. For wind-dispersed species

$$\bar{r} = s \cdot f_\sigma \cdot k, \tag{3}$$

where s is the wind strength, f_σ the spreading factor of the species and k the height class of the parent tree: seeds of taller trees fly farther. The direction of a wind-dispersed seed is determined in three steps:

1. Draw one of the eight wind directions with the frequencies of the wind rose (Table 2). Each direction covers a sector of $360°/8 = 45°$.
2. Draw the exact angle uniformly within this sector, e.g. for W between 247.5° and 292.5° (measured clockwise from north).
3. The wind rose gives the direction the wind comes *from*; the seed flies in the opposite direction (wind from W ⇒ seed flies towards E).

For animal-dispersed species the mean distance is a fixed value $\bar{r} = d_\sigma$ (in metres) and the direction is uniform. The seed lands in the grid cell nearest to its landing point. Seeds that land outside the grid are lost.

3. Establishment. Trees compete for light. Let $S \in [0,1]$ be the fraction of the eight neighbour cells (Moore neighbourhood) of a cell that hold a tree with a larger height class than the tree in this cell; for an empty cell, every neighbouring tree with $k \geq 1$ counts. Seeds that land on an occupied cell are lost. A seed of species σ on an empty cell establishes with probability

$$P_{\text{est}} = p_{\text{est}}\big(1 - (1 - \tau_\sigma)\, S\big), \tag{4}$$

where p_{est} is the establishment probability in full light and $\tau_\sigma \in [0,1]$ the shade tolerance of the species. If several seeds establish in the same cell, one of them is chosen uniformly at random. The new tree has age 0.

4. Growth. The age of every tree increases by one year.

5. Mortality. A tree of species σ dies with probability

$$p_{\text{die}} = \frac{1}{L_\sigma} + \beta\,(1 - \tau_\sigma)\, S. \tag{5}$$

The first term is the background mortality: L_σ is the mean life span of the species, so without shade a tree lives on average L_σ years. The second term is the mortality caused by shade, where S is the shade of the cell as defined above and τ_σ the shade tolerance of the species. The parameter $\beta \in [0,1]$ is the same for all species and sets the strength of the competition for light: it is the additional annual death probability of a completely shade-intolerant tree ($\tau_\sigma = 0$) under full shade ($S = 1$). For example, with $\beta = 0.3$ such a tree dies with an additional probability of 30 % per year, whereas a shade-tolerant tree with $\tau_\sigma = 0.8$ dies with only $0.3 \cdot (1 - 0.8) = 6\,\%$. While the shade acts on the establishment only once, when the seed lands, this term acts every year as long as a tree is smaller than its neighbours. Small trees under a canopy therefore die more often, while canopy trees are not affected by shade. For $\beta = 0$, shade has no effect on survival. Dead trees are removed and their cells become empty.

Default parameters. $\Delta = 5\,\text{m}$, $s = 10\,\text{m}$, $p_{\text{est}} = 0.5$, $\beta = 0.3$.

You are given an edited CSV file named `trees_vienna.csv` for Task 5.

Kind of tree	**Dispersal**	f / d **[m]**	φ	A **[yr]**	T **[yr]**	K	L **[yr]**	τ
Ahorn	wind	$f = 0.9$	2.0	20	12	5	120	0.80
Akazie	gravity	$d = 10$	3.0	6	7	5	60	0.18
Buche	animal	$d = 30$	1.2	30	18	6	150	0.89
Esche	wind	$f = 0.7$	2.4	20	9	6	100	0.42
Kastanie	animal	$d = 10$	0.8	20	14	5	100	0.61
Kiefer	wind	$f = 1.1$	2.0	15	10	5	100	0.22
Kirsche	animal	$d = 60$	2.4	8	8	3	50	0.58
Linde	wind	$f = 0.7$	1.6	25	14	6	150	0.80
Pappel	wind	$f = 3.0$	2.0	8	5	6	60	0.37
Platane	wind	$f = 1.8$	1.6	15	10	7	150	0.50
Zürgelbaum	animal	$d = 60$	1.6	15	14	4	100	0.54

Table 1: Species traits. f: spreading factor (wind dispersal), d: mean dispersal distance (animal or gravity dispersal), φ: fecundity factor, A: age at maturity, T: years per height class, K: maximal height class, L: mean life span, τ: shade tolerance.

N	**NE**	**E**	**SE**	**S**	**SW**	**W**	**NW**
6 %	5 %	6 %	18 %	7 %	6 %	30 %	22 %

Table 2: Wind rose for Vienna: relative frequency of the direction the wind comes from

To Do's

Task 1

Implement the seed dispersal for a single tree. Simulate the seed shadow (the two-dimensional distribution of landing positions) of every species for a tree of height class $k = 4$ and visualise it. Check that the mean simulated distance equals $\bar{r}$ and that the distances are exponentially distributed. Discuss how the wind rose shapes the seed shadow and how the species differ.

Task 2

Implement the agent-based model as defined above. Start with a random initial forest on a small grid. Create an animated visualisation of the model, and visually check that the model produces plausible results.

Task 3

Think about how to measure the outcome of the model quantitatively. Both the initial configuration and the dynamics involve randomness, so Monte Carlo simulation must be performed. Investigate at least the following two questions.

(a) How fast does a species colonise empty land? Relate it to the species traits.

(b) How does the species composition of a random initial forest develop over 250 years?

Task 4

Investigate the following questions by varying the model parameters (e.g. the wind strength s and the shade mortality β):

(a) Which species dominate after 50 years and which after 250 years? Compare their traits in Table 1: does dispersal or shade tolerance decide their success?

(b) Are there parameter values at which a small change leads to a different winning species?

(c) Seeds that land outside the grid are lost. How does this affect your results, and is it realistic?

Task 5

Use the tree register of Vienna as initialisation. You get it as a preprocessed list (`trees_vienna.csv`) in which every tree has a district, a position in metres east and north of a fixed reference point, one of the eleven species of Table 1 and a height class. Filter one district, for example the 4th district (Wieden), and convert the positions to grid cells. Think about the appropriate size of a cell and what happens if two trees are placed in the same cell. Simulate 200 years. Which species prevail after 25, 50, 100 and 200 years, and where?

Task 6

What could be done to make the model more realistic?